

Accelerated Observers and Born Rigidity

Sandro Vitenti

Scope

These notes conclude the discussion of accelerated observers in flat spacetime, in preparation for the study of radiation from moving charges.

The goals are:

- understand why Rindler coordinates are special;
- distinguish accelerated congruences from Born-rigid congruences;
- construct accelerated coordinates with identical proper acceleration.

Rindler Coordinates Revisited

Consider Minkowski spacetime with coordinates (T, X, Y, Z) and metric

$$ds^2 = -dT^2 + dX^2 + dY^2 + dZ^2.$$

Here $T \equiv x^0 = ct$ for ordinary time t , i.e. time measured in length units, consistent with the [conventions](#) fixed for this course; a proper acceleration α quoted below in these units corresponds to $\alpha_{\text{SI}} = c^2\alpha$ in SI units.

Define

$$T = \rho \sinh(a\eta), \quad X = \rho \cosh(a\eta).$$

Substituting,

$$ds^2 = -(a\rho)^2 d\eta^2 + d\rho^2 + dY^2 + dZ^2.$$

Worldlines $\rho = \text{const}$ satisfy

$$X^2 - T^2 = \rho^2.$$

These are hyperbolae.

Using proper time,

$$d\tau = a\rho d\eta,$$

the four-velocity is

$$u^\mu = \left(\cosh \frac{\tau}{\rho}, \sinh \frac{\tau}{\rho} \right),$$

and the four-acceleration is

$$a^\mu = \frac{1}{\rho} \left(\sinh \frac{\tau}{\rho}, \cosh \frac{\tau}{\rho} \right).$$

Hence

$$a^\mu a_\mu = \frac{1}{\rho^2}.$$

The proper acceleration is

$$\alpha = \frac{1}{\rho}.$$

i Note

Different Rindler observers have different proper accelerations. The observer closest to the horizon accelerates the most.

Born Rigidity

For a congruence with four-velocity u^μ , define

$$h_{\mu\nu} = \eta_{\mu\nu} + u_\mu u_\nu.$$

The spatial deformation tensor is

$$\Theta_{\mu\nu} = h_\mu{}^\alpha h_\nu{}^\beta \nabla_{(\alpha} u_{\beta)}.$$

Born rigidity requires

$$\Theta_{\mu\nu} = 0.$$

This means that neighboring observers maintain constant proper separation in their instantaneous rest frame. Rindler observers satisfy this condition.

Why Equal Acceleration is Incompatible with Born Rigidity

Suppose every observer had the same proper acceleration,

$$a^\mu a_\mu = a_0^2.$$

Imagine two neighboring observers separated by a fixed proper distance.

After a short interval, the rear observer must have acquired a slightly larger velocity than the front observer if the separation is to remain fixed in the instantaneous rest frame.

Therefore the proper acceleration must vary along the congruence.

The result is that translational Born-rigid acceleration necessarily requires an acceleration gradient.

Rindler coordinates realize exactly this behavior.

Accelerated Coordinates with Equal Proper Acceleration

Start from the uniformly accelerated trajectory

$$T_0(\tau) = \frac{1}{a} \sinh(a\tau), \quad X_0(\tau) = \frac{1}{a} \cosh(a\tau).$$

Translate the trajectory by a constant spatial label ξ :

$$T(\tau, \xi) = \frac{1}{a} \sinh(a\tau),$$

$$X(\tau, \xi) = \xi + \frac{1}{a} \cosh(a\tau).$$

Every worldline has

$$a^\mu a_\mu = a^2.$$

The different observers therefore possess identical proper acceleration.

Computing

$$dT = \cosh(a\tau) d\tau,$$

$$dX = d\xi + \sinh(a\tau) d\tau,$$

gives

$$ds^2 = -d\tau^2 + 2 \sinh(a\tau) d\tau d\xi + d\xi^2.$$

The metric has

$$g_{0i} \neq 0.$$

The congruence is accelerated but not of the Rindler type.

Comparison with Rindler

Property	Rindler	Equal-Acceleration Family
Proper acceleration	varies as $1/\rho$	constant
Born rigid	yes	no
Horizon	yes	no
g_{0i}	0	nonzero
Stationary	yes	no

Transition to Electrodynamics

Accelerated observers reveal nontrivial geometric effects. Accelerated charges reveal something more dramatic: they radiate. The energy-momentum tensor needed to make that statement precise, and the derivation of the radiation field itself, are developed in [Energy of the Electromagnetic Field](#) and [Radiation from a Moving Charge](#).

Summary

The Rindler congruence is a special Born-rigid accelerated frame whose proper acceleration varies spatially. Uniform proper acceleration for all observers can be achieved, but only at the cost of losing Born rigidity. These accelerated worldlines are exactly the ones whose radiation we study next.